Advance Reading

Knoll pp. 29-46, 112-115, 121-126, Ch. 7



Information

- Name:
- Lab Station:
- Lab Partner:
- Date:

1 Pre-Lab Conceptual Questions

1. What shape do you expect for the counting curve for a Geiger-Mueller Counter? Explain what causes each of the features.
2. What factors influence the height of pulses produced in a Geiger-Mueller Counter?
3. If you measured the gross counting G for the time interval t_g and the background counting B for time interval t_b, what is the error for the net count rate σ_r? Express σ_r in terms of G, B, t_g, and t_b.
4. Explain dead time, resolving time, and recovery time and how they are related in magnitude (which is longest, which is shortest). A sketch might be helpful.
5. What is the difference between paralyzable model and non-paralyzable model in dead time determination?

2 The Counting Curve

2.1 Equipment Setup

2.1.1 Required Equipment

- GM tube
- thorium source
- oscilloscope
- pre-amplifier
- amplifier
- single channel analyzer
- counter/timer
- multi channel analyzer